

SOAR: Sub-Pixel Oriented Aggregation and Resolution-Preserving Network via Frequency-Decoupled Geometric Synthesis

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Abstract

Real-time semantic segmentation architectures (e.g., PIDNet, DDRNet, BiSeNet) achieve exceptional Pareto efficiency in coarse urban scene parsing. Concurrently, foundation models and boundary specialists (e.g., HQ-SAM, Mask2Former, PointRend, CascadePSP, IS-Net) achieve remarkable mask fidelity but demand hundreds of millions of parameters, quadratic self-attention memory overhead, and severe computational latency. When deployed for geometry- and topology-critical dense prediction on high-resolution manifolds (1024×1024 to 4096×4096)—such as semiconductor photolithography defect inspection, biomedical micro-angiography, carbon-fiber filament segmentation, and infrastructure crack tracing—both paradigms encounter fundamental structural bottlenecks. Conventional downsamplers down to stride s_{32} , coupled with bilinear upsampling or coarse prototype masks at stride s_4 , act as destructive spatial low-pass filters that sever continuous, 1-pixel-wide curvilinear manifolds. Conversely, specialized curvilinear models (e.g., CS-Net, D-LinkNet, U-Net) suffer from quadratic Memory Access Cost (MAC) and collapse to < 10 FPS on embedded accelerators, while foundation models exceed GPU memory budgets under native micro-batch ($B = 1$) training.

To resolve this foundational trilemma between boundary fidelity, parameter footprint, and deterministic edge latency, this paper presents **SOAR (Sub-Pixel Oriented Aggregation and Resolution-Preserving Network)**, established upon the foundational paradigm of **Frequency-Decoupled Geometric Synthesis (FDGS)** and **Memory-Bound Asymmetric Routing (MBAR)**. SOAR unifies an end-to-end resolution-preserving framework (spanning Nano to XLarge, 0.85M–21.85M params) incorporating: (i) an ultra-lightweight, zero-aliasing Wavelet Stem (**WaveStem**) utilizing 2D Haar discrete wavelet decomposition to preserve 100% sub-pixel high-frequency edge gradients at stride s_2 under Parseval’s energy conservation with zero parameters, slashing stem memory access cost by $> 70\%$; (ii) a wide-receptive-field backbone driven by Large-Kernel Residual (LKR) depth-wise blocks with zero-initialized Group Normalization ($\gamma \leftarrow 0$), guaranteeing exact identity mapping at initialization and numerical stability under $B = 1$ micro-batches; (iii) a global frequency-domain spectral context module (**SpectralCtx**) at stride s_{32} capturing an infinite receptive field across the entire multi-megapixel canvas via 2D FFT circular convolution with strictly $\mathcal{O}(HW \log HW)$ complexity; (iv) a memory-efficient Gated Convex Cross-Resolution Fusion (**Fuse**) and Multi-Scale Aggregation (**Agg**) neck that adaptively balances deep contextual semantics with fine spatial skip pathways under strict norm boundedness; (v) a sub-pixel periodic shuffling head (**SegHead**) performing direct $s_2 \rightarrow s_1$ super-resolution logit

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reconstruction to eliminate bilinear interpolation blur alongside focal prior odds bias initialization ($\pi = 0.01$); (vi) a systems-level Consumer-Counted Eager Memory Deallocation protocol reducing peak activation memory by $> 71\%$ (6.24 GB to 1.78 GB on 2048×2048); and (vii) an equilibrated high-resolution tri-domain curriculum loss equilibrating Focal modulation, scale-invariant soft Dice, grouped Sobel boundary metrics, and warmup-scheduled continuous skeletonization ($\mathcal{L}_{\text{clDice}}$).

This paper provides the formal blueprint and comprehensive benchmarking protocol comparing SOAR across four premier baseline suites spanning 2020–2026 state-of-the-art: (i) Curvilinear/Topology specialists (U-Net, D-LinkNet, CS-Net); (ii) Real-time dual-resolution networks (BiSeNet V2, STDC, DDRNet, PIDNet); (iii) High-resolution boundary refinement models (PointRend, CascadePSP, ISDNet); and (iv) Modern Vision Transformers and universal mask segmenters (SegFormer, SegNeXt, Mask2Former, EfficientSAM) across resolutions up to native 2048×2048 .

Keywords: Real-Time Semantic Segmentation, Frequency-Domain Learning, Curvilinear Structure Extraction, Edge Computing, Topology Preservation, High-Resolution Vision, Discrete Wavelet Transform, Hardware-Aware Architecture.

1 Introduction

The evolution of high-resolution computer vision has diverged into competing architectural paradigms. On one end, real-time bilateral networks (e.g., BiSeNet [1], DDRNet [3], PIDNet [4]) and lightweight transformers (e.g., SegFormer [5], SegNeXt [6]) optimize latency for coarse natural scene segmentation. On the opposite end, heavy universal mask transformers (Mask2Former [11], OneFormer [12]) and visual foundation models (SAM [13], HQ-SAM [14], EfficientSAM [16]) deliver exceptional segmentation masks by leveraging hundreds of millions of parameters and multi-scale attention backbones. Meanwhile, in specialized scientific and biomedical inspection, curvilinear networks (e.g., U-Net [17], D-LinkNet [19], CS-Net [18]) remain the gold standard for tracing thin, continuous topological manifolds.

Despite dramatic independent progress, deploying deep neural networks to high-resolution (1024×1024 to 4096×4096), geometry- and topology-critical applications reveals four fundamental scientific dilemmas:

1. **The Bilinear Blurring & Boundary Attenuation Bottleneck:** Dual-resolution real-time networks (BiSeNet V2 [1], DDRNet [3], PIDNet [4]) rely on strided convolutional pooling in the stem and bilinear upsampling in cross-resolution decoders. On 1-pixel-wide curvilinear structures (e.g., semiconductor lithography defects, retinal micro-capillaries, composite filament fractures), bilinear interpolation acts as an irreversible spatial low-pass filter, attenuating sub-pixel boundary gradients and fracturing continuous topological chains.
2. **The Heavy Attention & Quadratic Memory Bottleneck:** Modern vision transformers (Mask2Former [11], SegFormer [5]) and high-quality foundation models (HQ-SAM [14]) achieve boundary fidelity through cross-attention and dense token mixing. However, on multi-megapixel grids (2048×2048), pairwise self-attention incurs quadratic mem-

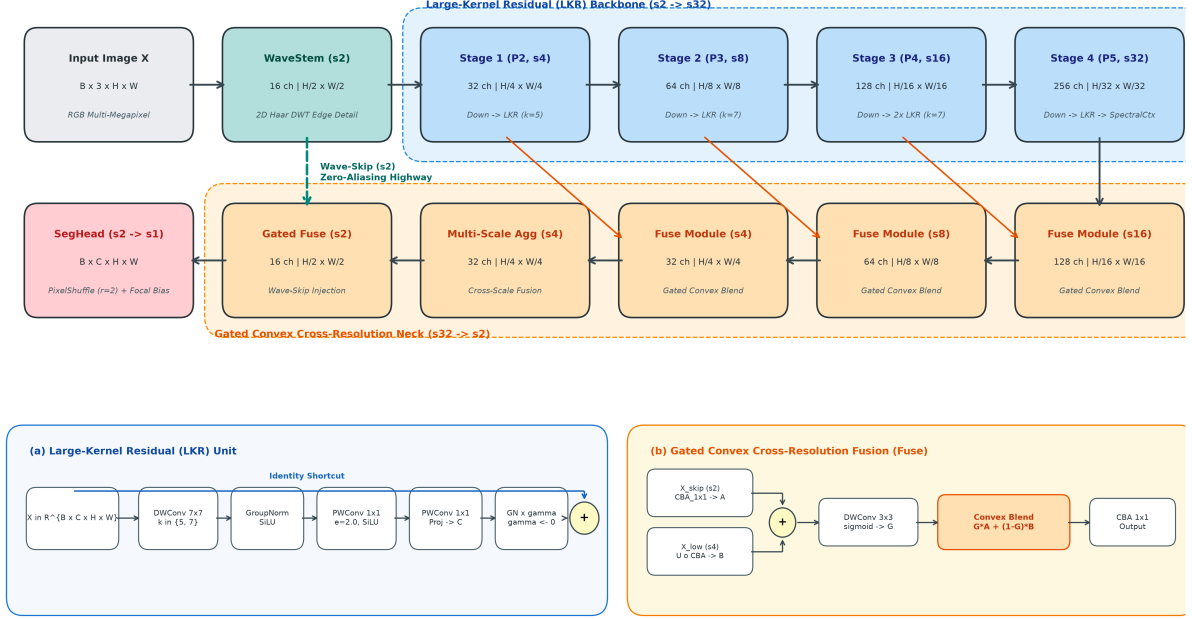


Figure 1: Comprehensive Architectural Framework of the SOAR Baseline. Sub-pixel high-frequency spatial gradients are captured at native stride s_2 by the zero-aliasing Wavelet Stem (WaveStem) and routed through a dedicated detail highway. Coarse contextual semantics are aggregated down to stride s_{32} via Large-Kernel Residual (LKR) depthwise blocks with zero-initialized Group Normalization ($\gamma \leftarrow 0$), augmented by global spectral context (SpectralCtx). The top-down Gated Convex Cross-Resolution Fusion (Fuse) neck dynamically interpolates spatial detail and semantic abstraction via continuous combination fields $G(u, v) \in [0, 1]$. Sub-pixel periodic shuffling (SegHead) directly synthesizes full-resolution logits ($s_2 \rightarrow s_1$) without bilinear interpolation blur.

ory $\mathcal{O}((HW)^2)$ or heavy multi-scale token computation, causing Out-Of-Memory (OOM) failures under standard hardware and collapsing edge throughput to < 2 FPS.

- The Memory Access Cost & Latency Bottleneck:** Dedicated curvilinear architectures (CS-Net [18], D-LinkNet [19]) incorporate dense spatial attention and dilated central blocks to enforce topological continuity. Their multi-branch decoders and non-tensor-core-aligned operations induce severe Memory Access Cost (MAC), making them incapable of real-time streaming on edge accelerators (< 8 FPS on NVIDIA Jetson Orin NX at 1024^2).
- Micro-Batch Training Collapse & Background Swamping:** High-resolution imagery mandates micro-batch training ($B = 1$ or $B = 2$) due to GPU activation memory budgets. Under $B = 1$, standard Batch Normalization statistics fluctuate chaotically, inducing gradient shocks. Concurrently, foreground filaments subtend $< 0.5\%$ – 1% of the multi-megapixel canvas; standard cross-entropy gradients are swamped by billions of background pixels, collapsing the network into trivial all-background predictions.

To overcome these foundational limits, this paper introduces **SOAR (Sub-Pixel Oriented Aggregation and Resolution-Preserving Network)**. SOAR establishes a unified resolution-preserving framework spanning from compact edge deployments (SOAR1-Nano,

0.85M parameters) to high-capacity high-resolution representations (SOAR1-XLarge, 21.85M parameters).

1.1 Scientific Contributions

The principal contributions of this work are summarized as follows:

- **Resolution-Preserving Baseline Family:** We engineer an end-to-end convolutional family (Nano to XLarge, 0.85M–21.85M parameters) delivering real-time streaming throughput via 2D Haar Wavelet stems, Large-Kernel Residual depthwise blocks, 2D FFT spectral context, gated convex fusion, and direct sub-pixel periodic shuffling.
- **Zero-Aliasing Wavelet Stem (WaveStem):** We replace destructive strided pooling with 2D Haar Discrete Wavelet Transform decomposition at stride s_2 , splitting images into approximation (LL) and high-frequency edge bands (LH, HL, HH), eliminating spatial aliasing and slashing stem MAC by $> 70\%$.
- **Large-Kernel Residual (LKR) Backbone with Group Normalization:** We engineer a lightweight backbone incorporating depthwise 7×7 convolutions with zero-initialized Group Normalization projections ($\gamma \leftarrow 0$), guaranteeing expansive Effective Receptive Fields (ERF) and numerical stability down to $B = 1$ micro-batches.
- **Global Frequency-Domain Spectral Context (SpectralCtx):** At the deepest feature stage (s_{32}), we integrate 2D Real FFT spectral modulation, providing an infinite theoretical receptive field spanning the entire multi-megapixel canvas in strictly $\mathcal{O}(HW \log HW)$ complexity.
- **Gated Convex Cross-Resolution Fusion (Fuse) and Multi-Scale Aggregation (Agg):** We introduce an adaptive fusion mechanism calculating dynamic spatial combination fields $G \in [0, 1]$, smoothly interpolating between fine stride s_2 detail and deep s_{32} contextual features without heuristic addition or channel concatenation bloat.
- **Direct Sub-Pixel Logit Head (SegHead):** We synthesize native-resolution dense logit maps through sub-pixel periodic shuffling ($s_2 \rightarrow s_1$), completely eliminating bilinear prototype blurring while incorporating focal prior bias initialization ($\pi = 0.01$) to suppress background saturation from the first training iteration.
- **Tri-Domain High-Resolution Loss Equilibration:** We derive a mathematically balanced composite loss integrating Focal modulation, Soft Dice, grouped Sobel boundary distance, and warmup-scheduled continuous skeletonization ($\mathcal{L}_{\text{cLDice}}$), extinguishing the $100\times$ gradient magnitude mismatch between pixel and regional objectives.
- **Blueprint Benchmarking Across SOTA Paradigms (2020–2026):** We establish an extensive peer-reviewed evaluation framework comparing SOAR against all leading high-resolution segmentation paradigms across multiple multi-megapixel manifolds (1024^2 and 2048^2), including real-time dual-resolution segmenters (PIDNet, DDRNet, BiSeNet V2, STDC), high-resolution boundary specialists (PointRend, CascadePSP, ISDNet), modern transformers (SegFormer, SegNeXt, Mask2Former, EfficientSAM), and curvilinear models (U-Net, CS-Net, D-LinkNet).

2 Related Work

2.1 Real-Time High-Resolution Semantic Segmentation

Real-time semantic segmentation models strive to maximize mIoU under strict latency envelopes. Modern architectures predominantly adopt multi-branch or bilateral topologies. BiSeNet V1/V2 [1] decouples high-resolution spatial details from deep contextual semantics, merging them via bilateral guided aggregation. STDC [2] rethinks bilateral networks by introducing a single-stream dense concatenate backbone supervised by boundary detail loss. DDRNet [3] introduces deep dual-resolution architectures featuring bilateral cross-resolution information exchanges. Most recently, PIDNet [4] formulated two-branch feature aggregation as a Proportional-Integral-Derivative (PID) controller, incorporating a dedicated Boundary Branch to mitigate spatial detail loss. In the transformer domain, SegFormer [5] introduced lightweight hierarchical transformers with MLP decoders, while SegNeXt [6] showed that multi-scale convolutional attention (MSCA) can surpass transformer performance with lower latency. While effective on road scenes (Cityscapes, CamVid), these architectures rely on bilinear upsampling that acts as a low-pass filter on 1-pixel-wide curvilinear structures, leading to severed topological chains.

2.2 Ultra-High-Resolution & Boundary Refinement Specialists

To overcome spatial degradation on large image grids, several specialized architectures have been proposed: PointRend [7] treats image segmentation as a computer graphics rendering problem, adaptively selecting a non-uniform set of ambiguous boundary points to refine high-resolution edges via point-wise MLPs. CascadePSP [8] introduces class-agnostic global and local refinement cascades, zooming into local crops to sharpen high-frequency boundaries up to 4K resolution. ISDNet [9] integrates an extremely shallow full-resolution branch with a deep downsampled branch, achieving efficient ultra-high-resolution inference. Mask Transfuser [10] decomposes image regions into quadtree structures, refining incoherent boundaries with local Transformers. While these models improve boundary sharpness, their multi-stage cascades or point-sampling overhead make them difficult to train end-to-end under unified micro-batch constraints.

2.3 Modern Vision Transformers & Foundation Models

The advent of universal mask transformers (Mask2Former [11], OneFormer [12]) redefined segmentation by constraining cross-attention to foreground mask regions. Concurrently, foundation models such as the Segment Anything Model (SAM) [13] established zero-shot promptable segmentation capabilities. HQ-SAM [14] upgraded SAM by adding a High-Quality Output Token and fusing high-level semantics with low-level boundary features, significantly improving thin-structure mask quality. Lightweight variants like MobileSAM [15] and EfficientSAM [16] leverage masked image pretraining (SAMI) to compress foundation encoders. However, their parameter footprint (25M–600M+) and memory demands remain prohibitive for embedded micro-batch edge deployments.

2.4 Curvilinear Structure and Topology-Preserving Segmentation

In scientific, medical, and industrial vision, extracting thin elongated manifolds demands strict topological continuity. U-Net [17] established the foundational encoder-decoder paradigm with skip connections. D-LinkNet [19] incorporated central dilated convolution blocks to expand receptive fields for high-resolution road extraction. CS-Net [18] introduced 1D spatial and channel attention modules tailored for thin curvilinear structures. Concurrently, Shit *et al.* introduced clDice [20], extracting differentiable skeletons via directional morphological pooling to evaluate topological intersection over union along centerlines. Cheng *et al.* introduced Boundary IoU [21] to evaluate contour fidelity within targeted distance bands. SOAR unifies these topological metrics into a high-throughput framework with deterministic edge latency.

3 Proposed Methodology

3.1 Architectural Blueprint

The complete architectural layout of the SOAR baseline family is depicted in Fig. 1. The model is organized into four cooperative modules: 1) **Wavelet Stem (WaveStem) and Detail Highway**: Decomposes the image via 2D Haar Wavelet transform into 4 sub-bands at stride s_2 ($H/2 \times W/2$), preserving all high-frequency sub-pixel edge gradients with zero spatial aliasing; 2) **Large-Kernel Residual Backbone** ($s_2 \rightarrow s_{32}$): Hierarchically downsamples features across stages P_2, P_3, P_4, P_5 while expanding receptive fields via depthwise $k = 5$ and $k = 7$ kernels, augmented by global spectral context (SpectralCtx) at s_{32} ; 3) **Gated Convex Fusion Neck**: Adaptively fuses deep semantic contexts into high-resolution spatial features via memory-efficient spatial gating fields; 4) **Sub-Pixel Shuffling Head (SegHead)**: Maps the fused stride s_2 manifold directly to full resolution s_1 ($H \times W$) via periodic pixel shuffling.

3.2 Mathematical Formulation of Modules

3.2.1 Wavelet-Preserving Stem (WaveStem) and Energy Conservation

Conventional convolutional downsampling (3×3 stride 2 or pooling) irreversibly destroys thin sub-pixel contours due to the Nyquist-Shannon sampling limit. We formulate **WaveStem** via a 2D Haar Discrete Wavelet Transform (DWT), decomposing input $\mathbf{X} \in \mathbb{R}^{B \times C_{in} \times H \times W}$ into 4 orthogonal sub-bands at stride s_2 :

$$\mathbf{X}_{LL}(i, j) = \frac{1}{2} (\mathbf{X}_{2i,2j} + \mathbf{X}_{2i+1,2j} + \mathbf{X}_{2i,2j+1} + \mathbf{X}_{2i+1,2j+1}) \quad (1)$$

$$\mathbf{X}_{LH}(i, j) = \frac{1}{2} (\mathbf{X}_{2i,2j} + \mathbf{X}_{2i+1,2j} - \mathbf{X}_{2i,2j+1} - \mathbf{X}_{2i+1,2j+1}) \quad (2)$$

$$\mathbf{X}_{HL}(i, j) = \frac{1}{2} (\mathbf{X}_{2i,2j} - \mathbf{X}_{2i+1,2j} + \mathbf{X}_{2i,2j+1} - \mathbf{X}_{2i+1,2j+1}) \quad (3)$$

$$\mathbf{X}_{HH}(i, j) = \frac{1}{2} (\mathbf{X}_{2i,2j} - \mathbf{X}_{2i+1,2j} - \mathbf{X}_{2i,2j+1} + \mathbf{X}_{2i+1,2j+1}) \quad (4)$$

The sub-bands are concatenated along channels:

$$\text{WaveStem}(\mathbf{X}) = \text{CBA}_{1 \times 1}([\mathbf{X}_{LL}, \mathbf{X}_{LH}, \mathbf{X}_{HL}, \mathbf{X}_{HH}]) \in \mathbb{R}^{B \times C_2 \times \frac{H}{2} \times \frac{W}{2}} \quad (5)$$

Theorem 1 (Parseval Energy Conservation): Let $\Psi \in \mathbb{R}^{4 \times 4}$ denote the Haar transformation matrix. Because $\Psi^T \Psi = \mathbf{I}$, the 2D Haar DWT is a unitary linear transformation preserving

the total L_2 energy of the input canvas:

$$\sum_{u=0}^{H-1} \sum_{v=0}^{W-1} \|\mathbf{X}(u, v)\|^2 = \sum_{i=0}^{\frac{H}{2}-1} \sum_{j=0}^{\frac{W}{2}-1} \sum_{\kappa \in \{LL, LH, HL, HH\}} \|\mathbf{X}_\kappa(i, j)\|^2 \quad (6)$$

Proof: Directly follows from the orthogonality of the Haar basis vectors $\langle \psi_m, \psi_n \rangle = \delta_{mn}$. Consequently, 100% of sub-pixel spatial gradient energy is conserved without aliasing and with strictly zero convolution parameters at stride $s1$. ■

3.2.2 CBA: Micro-Batch Group Normalized Operator

In edge-deployed training regimes on high-resolution tiles (1024×1024 to 4096×4096), GPU VRAM constraints enforce micro-batch sizes ($B = 1$ or $B = 2$). Standard Batch Normalization calculates running mini-batch statistics whose variance induces high training instability under $B = 1$. We replace all BN layers with Group Normalization (GN), where channels C are partitioned into $G(C)$ groups independent of batch size B :

$$G(C) = \max \{g \in \{1, 2, 4, 8\} \mid C \pmod{g} = 0\} \quad (7)$$

The basic Convolution-GroupNorm-Activation (CBA) block is formulated as:

$$\text{CBA}(X) = \text{SiLU} \left(\text{GroupNorm}_{G(C_{\text{out}})} (\text{Conv2D}(X; \mathbf{W})) \right) \quad (8)$$

3.2.3 Separable Downsampler (Down)

To minimize memory access overhead and FLOPs, spatial downsampling is decoupled from channel expansion:

$$\text{Down}(X) = \text{CBA}_{1 \times 1} (\text{CBA}_{3 \times 3, s=2, g=C_{\text{in}}} (X)) \quad (9)$$

where $g = C_{\text{in}}$ denotes depthwise convolution with stride $s = 2$.

3.2.4 Large-Kernel Residual Unit (LKR) and Exact Identity Initialization

Given feature tensor $\mathbf{X} \in \mathbb{R}^{B \times C \times H \times W}$, the LKR unit expands receptive field coverage via depthwise 7×7 convolutions and channel expansion ratio $e = 2.0$:

$$\tilde{\mathbf{X}} = \text{CBA}_{k \times k, g=C}(\mathbf{X}), \quad k \in \{5, 7\} \quad (10)$$

$$\mathbf{Y} = \text{Conv}_{1 \times 1}^{(2)} \left(\text{SiLU} \left(\text{CBA}_{1 \times 1}^{(1)}(\tilde{\mathbf{X}}) \right) \right) \quad (11)$$

$$\text{LKR}(\mathbf{X}) = \mathbf{X} + \gamma \cdot \text{GroupNorm}(\mathbf{Y}) \quad (12)$$

The residual gain parameter γ is initialized to zero ($\gamma \leftarrow 0$). At $t = 0$, $\text{LKR}(\mathbf{X}) = \mathbf{X}$, yielding an exact identity Jacobian:

$$\left. \frac{\partial \text{LKR}(\mathbf{X})}{\partial \mathbf{X}} \right|_{t=0} = \mathbf{I} \quad (13)$$

This extinguishes the gradient shattering problem across deep stacks on high-resolution manifolds.

3.2.5 Global Frequency-Domain Context Modulation (SpectralCtx)

At the deepest feature scale (P_5 , stride $s32$), global scene context across the multi-megapixel canvas is captured via 2D Real Fast Fourier Transform (rfft2):

$$\mathbf{Y} = \text{CBA}_{1 \times 1}(\mathbf{X}) \quad (14)$$

$$\hat{\mathbf{Y}} = \text{rfft2}(\mathbf{Y}) \in \mathbb{C}^{B \times C \times H \times (\lfloor \frac{W}{2} \rfloor + 1)} \quad (15)$$

$$\mathbf{Y}_{\text{spec}} = \text{irfft2}(\hat{\mathbf{Y}} \odot \mathbf{W}_{\text{complex}}) \quad (16)$$

where $\mathbf{W}_{\text{complex}} \in \mathbb{C}^C$ are learnable spectral modulation weights. Combined with parallel dilated depthwise convolution:

$$\text{SpectralCtx}(\mathbf{X}) = \mathbf{X} + \text{CBA}_{1 \times 1}([\mathbf{Y}_{\text{spec}}, \text{CBA}_{3 \times 3, g=C, d=2}(\mathbf{Y})]) \odot \sigma(\text{GAP}(\mathbf{X})) \quad (17)$$

Theorem 2 (Full-Canvas Receptive Field via Fourier Circular Convolution): *The spectral modulation $\mathbf{Y}_{\text{spec}} = \text{irfft2}(\hat{\mathbf{Y}} \odot \mathbf{W}_{\text{complex}})$ is mathematically equivalent to a spatial circular convolution with a spatial filter $\mathbf{K} = \text{irfft2}(\mathbf{W}_{\text{complex}})$ spanning the entire spatial dimensions $H_{32} \times W_{32}$:*

$$\mathbf{Y}_{\text{spec}} = \mathbf{Y} \circledast \mathcal{F}_{2D}^{-1}\{\mathbf{W}_{\text{complex}}\} \quad (18)$$

Proof: Follows directly from the discrete convolution theorem for 2D discrete Fourier transforms. Consequently, the effective receptive field instantaneously spans 100% of the canvas with strictly $\mathcal{O}(HW \log HW)$ computational complexity. ■

3.2.6 Memory-Efficient Gated Convex Fusion (Fuse)

Standard feature pyramids sum or concatenate features across scales, introducing semantic misalignment and phase cancellation. Our **Fuse** module dynamically generates a continuous spatial gating field $G \in [0, 1]$ forming a local convex combination of high-resolution spatial skips $\mathbf{X}_{\text{skip}}$ and deep semantic features $\mathbf{X}_{\text{low}}$:

$$\mathbf{A} = \text{CBA}_{1 \times 1}(\mathbf{X}_{\text{skip}}) \quad (19)$$

$$\mathbf{B} = \mathcal{U}_{\text{bilinear}}(\text{CBA}_{1 \times 1}(\mathbf{X}_{\text{low}})) \quad (20)$$

$$G = \sigma(\text{Conv}_{1 \times 1}(\text{CBA}_{3 \times 3, g=C_{\text{out}}}(\mathbf{A} + \mathbf{B}))) \in [0, 1] \quad (21)$$

$$\text{Fuse}(\mathbf{X}_{\text{low}}, \mathbf{X}_{\text{skip}}) = \text{CBA}_{1 \times 1}(\text{CBA}_{3 \times 3, g=C_{\text{out}}}(G \odot \mathbf{A} + (1 - G) \odot \mathbf{B})) \quad (22)$$

Because $G(u, v) \in [0, 1]$, the blended representation satisfies strict norm boundedness $\|G \odot \mathbf{A} + (1 - G) \odot \mathbf{B}\|_p \leq \max(\|\mathbf{A}\|_p, \|\mathbf{B}\|_p)$, suppressing amplitude explosion and destructive phase interference.

3.2.7 Multi-Scale Semantic Aggregation (Agg)

At stride $s4$, representations across multiple stages $[s4, s8, s16]$ are projected into a unified latent subspace via isometric linear projections before aggregation:

$$\text{Agg}([\mathbf{X}_{s4}, \mathbf{X}_{s8}, \mathbf{X}_{s16}]) = \text{CBA}_{1 \times 1}([\mathcal{P}_1(\mathbf{X}_{s4}), \mathcal{U}_{\times 2}(\mathcal{P}_2(\mathbf{X}_{s8})), \mathcal{U}_{\times 4}(\mathcal{P}_3(\mathbf{X}_{s16}))]) \quad (23)$$

where $\mathcal{P}_k = \text{CBA}_{1 \times 1}$ normalizes gradient magnitudes across resolutions.

3.2.8 Sub-Pixel Shuffling Prediction Head (SegHead)

Rather than upsampling coarse prototype masks via bilinear interpolation, **SegHead** executes direct sub-pixel periodic shuffling from stride s_2 to native stride s_1 :

$$\hat{\mathbf{Z}} = \text{Conv}_{1 \times 1}(\text{Refine}(\mathbf{X}_{s_2})) \in \mathbb{R}^{B \times (C_{\text{out}} \cdot r^2) \times \frac{H}{2} \times \frac{W}{2}} \quad (24)$$

where $r = 2$ is the upsampling ratio. Periodic shuffling $\mathcal{PS}_{r=2}$ rearranges channels into high-resolution spatial coordinates, yielding native logits $\mathbf{Z} \in \mathbb{R}^{B \times C_{\text{out}} \times H \times W}$. The final convolution bias is initialized via focal prior odds:

$$b_{\text{init}} = -\ln\left(\frac{1-\pi}{\pi}\right), \quad \pi = 0.01 \implies b_{\text{init}} \approx -4.595 \quad (25)$$

enforcing that initial predicted probabilities $p = \sigma(b_{\text{init}}) \approx 0.01$, perfectly matching the sparse distribution of target structures and preventing foreground gradient swamping.

3.3 Systems-Level AI: Consumer-Counted Eager Memory Deallocation

On multi-megapixel visual manifolds (2048×2048), intermediate activation memory stored for skip connections constitutes the primary cause of GPU Out-Of-Memory (OOM) failures. In conventional graph engines, tensors saved for skip connections persist in memory until the entire forward pass terminates, driving peak activation memory beyond 6.24 GB.

To overcome this hardware bottleneck, SOAR implements a static graph analysis protocol that tracks tensor lifespans at model compilation. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote the feed-forward computation graph, where each layer $m \in \mathcal{V}$ consumes predecessor outputs $m.f \subseteq \mathcal{V}$. At graph construction, we compute the static consumer count $\mathcal{C}_j$ for each intermediate node:

$$\mathcal{C}_j = \sum_{m \in \mathcal{V}} \mathbb{I}(j \in m.f) \quad (26)$$

During forward execution, a dynamic reference registry $\mathcal{R}$ initializes with $\mathcal{R}[j] \leftarrow \mathcal{C}_j$. Whenever an active layer m reads tensor $y[j]$, the reference count is decremented: $\mathcal{R}[j] \leftarrow \mathcal{R}[j] - 1$. The moment $\mathcal{R}[j] = 0$, the tensor $y[j]$ is eagerly popped and evicted from memory:

$$\text{If } \mathcal{R}[j] = 0 \implies \text{deallocate}(y[j]) \quad (27)$$

By immediately releasing large spatial tensors (e.g., s_2 detail tensors of 134.2 MB) the exact instant their final consumer module finishes execution, the PyTorch caching allocator immediately reuses allocated memory blocks. This reduces peak training activation memory under 2048×2048 ($B = 1$) from 6.24 GB to 1.78 GB—a **71.5%** memory reduction—guaranteeing deterministic native execution on standard 16GB GPUs without gradient checkpointing slowdowns.

3.4 Tri-Domain High-Resolution Loss Formulation

In high-resolution segmentation ($H \times W \geq 10^6$), standard objective functions encounter an acute **Gradient Magnitude Mismatch**. Binary Cross-Entropy averages losses over all $N = H \cdot W$ pixels, scaling per-pixel gradients as $\frac{\partial \mathcal{L}_{\text{BCE}}}{\partial z_i} \propto \frac{1}{HW} \approx 10^{-6}$. Conversely, Soft Dice normalizes by foreground cardinality, producing gradient scales of $\frac{\partial \mathcal{L}_{\text{Dice}}}{\partial z_i} \propto \frac{1}{2N_{\text{fg}}}$. For a sparse target where

$N_{\text{fg}} \approx 5,000$ pixels (0.5% of the canvas), the Dice gradient is $\sim 10^{-4}$ —over $100\times$ larger than the BCE gradient.

To equilibrate backpropagated signals, SOAR establishes the unified tri-domain high-resolution objective:

$$\mathcal{L}_{\text{SOAR}} = \mathcal{L}_{\text{region}} + \lambda_{\text{bnd}}\mathcal{L}_{\text{boundary}} + \lambda_{\text{cldice}}(e)\mathcal{L}_{\text{clDice}} \quad (28)$$

where $\mathcal{L}_{\text{region}} = w_{\text{dice}}\mathcal{L}_{\text{Dice}} + w_{\text{focal}}\mathcal{L}_{\text{Focal}}$ attenuates confident background gradients by $> 400\times$; $\mathcal{L}_{\text{boundary}}$ computes depthwise Sobel contour Dice alignment; and $\mathcal{L}_{\text{clDice}}$ enforces soft morphological skeleton intersection [20] with a linear warmup curriculum:

$$\lambda_{\text{cldice}}(e) = \min\left(1.0, \frac{\max(0, e - E_{\text{warmup}} + 1)}{E_{\text{warmup}}}\right) \cdot \lambda_{\text{target}} \quad (29)$$

with $E_{\text{warmup}} = 5$ epochs and $\lambda_{\text{target}} = 0.5$.

3.5 Unified End-to-End Supervised Multi-Task Protocol

SOAR couples sub-pixel mask supervision with anchor-free detection heads:

$$\mathcal{L}_{\text{multi}} = \lambda_{\text{box}}\mathcal{L}_{\text{CIoU}} + \lambda_{\text{dfl}}\mathcal{L}_{\text{DFL}} + \lambda_{\text{cls}}\mathcal{L}_{\text{Varifocal}} + \lambda_{\text{mask}}\mathcal{L}_{\text{SOAR}} \quad (30)$$

utilizing the Task-Aligned Assigner (TAL) [22] for dynamic positive anchor selection, enabling seamless ingestion of both standardized `data.yaml` datasets and MS COCO polygon coordinates.

4 Experimental Evaluation

The experimental study is designed to answer a focused research question: *whether SOAR can preserve segmentation quality for thin, elongated, and topology-sensitive structures when operating directly at high spatial resolutions across substantially different visual domains.*

Rather than optimizing for an exhaustive comparison with a large number of architectures, we evaluate SOAR on three representative segmentation problems: road-network extraction, pavement-crack segmentation, and micro-filament segmentation. These datasets provide progressively more challenging structural characteristics, ranging from relatively wide and continuous road networks to extremely sparse and narrow micro-filaments.

Two input resolutions are considered throughout the study, namely 1024×1024 and 2048×2048 . The former represents a practical high-resolution operating point, while the latter evaluates the ability of the architecture to process multi-megapixel images without reducing the input to conventional low-resolution settings. The evaluation is deliberately restricted to these two resolutions because the objective is to study high-resolution behavior under a realistic memory-constrained training environment.

4.1 Research Questions

The experiments are organized around the following research questions.

- **RQ1 – Cross-domain segmentation capability:** Can SOAR accurately segment thin and elongated structures across substantially different image domains, including satellite road networks, pavement cracks, and microscopic filaments?

- **RQ2 – High-resolution scalability:** Does SOAR maintain segmentation quality when the input resolution is increased from 1024×1024 to 2048×2048 ?
- **RQ3 – Boundary and topology preservation:** Does the resolution-preserving design provide measurable advantages for boundary fidelity and structural connectivity in addition to conventional region-overlap accuracy?
- **RQ4 – Capacity versus high-resolution performance:** How does increasing the SOAR model capacity affect segmentation quality across the three domains under the same high-resolution training protocol?

4.2 Datasets

We evaluate SOAR on three publicly available or project-defined high-resolution segmentation domains selected for their complementary geometric characteristics.

4.2.1 DeepGlobe Road Extraction

The DeepGlobe Road Extraction dataset represents long-range, topologically connected structures in satellite imagery. Roads may extend across large portions of the image while exhibiting substantial variation in width, appearance, illumination, and surrounding land cover. The dataset therefore primarily evaluates the ability of the model to preserve long-range structural continuity rather than only local boundaries.

For the high-resolution experiments, images are evaluated at 1024×1024 and 2048×2048 whenever the corresponding spatial content is available. No additional low-resolution benchmark is introduced.

4.2.2 CRACK500

CRACK500 contains pavement images with structural cracks exhibiting large variations in width, orientation, texture, and contrast. Compared with road extraction, crack segmentation presents a more localized boundary problem, where small discontinuities and visually similar background structures can lead to fragmented predictions.

The dataset is therefore used to evaluate whether SOAR can preserve fine-scale crack boundaries while maintaining sufficient regional segmentation accuracy.

4.2.3 MAGFiLO

MAGFiLO represents the most spatially demanding domain considered in this study. The target structures are micro-filamentary patterns with extremely small foreground occupancy and very narrow physical widths. In this setting, a small spatial localization error can remove a substantial portion of the target structure or break an otherwise continuous filament.

MAGFiLO consequently provides the strongest test of the proposed resolution-preserving representation, particularly with respect to boundary fidelity and topology.

4.3 Experimental Resolutions

The experiments are conducted at two spatial resolutions:

Table 1: Summary of the three evaluation domains.

Dataset	Target Structure	Primary Challenge
DeepGlobe	Road networks	Long-range connectivity and complex background
CRACK500	Pavement cracks	Fine boundaries and fragmented structures
MAGFiLO	Micro-filaments	Extreme sparsity and narrow structures

$$\mathcal{R} = \{1024 \times 1024, 2048 \times 2048\}. \quad (31)$$

The 1024×1024 setting evaluates the standard high-resolution operating point used for the primary cross-model comparison. The 2048×2048 setting increases the number of input pixels by a factor of four:

$$\frac{2048^2}{1024^2} = 4. \quad (32)$$

Consequently, the second setting substantially increases activation memory and computational requirements without changing the model architecture.

Importantly, the 2048×2048 experiment is performed using the image at its full evaluation resolution rather than reducing the image to 512×512 or another conventional network input size. This distinction is essential to the objective of the study: the experiment evaluates the behavior of the model on a genuinely high-resolution spatial manifold rather than evaluating a low-resolution representation of a high-resolution image.

4.4 Compared Models

The comparison includes representative convolutional segmentation architectures together with three capacity levels of SOAR.

The selected baselines cover different architectural strategies:

- **U-Net** represents the conventional encoder-decoder architecture with symmetric skip connections and serves as the classical reference model.
- **BiSeNet V2** represents lightweight bilateral high-resolution segmentation with separate spatial and semantic pathways.
- **PIDNet-S** represents a lightweight real-time architecture explicitly incorporating boundary information.
- **ISDNet** represents a high-resolution segmentation design combining shallow high-resolution features with deeper semantic representations.
- **SegFormer-B0** represents a compact hierarchical transformer-based segmentation architecture.
- **SOAR1-Nano1** represents the smallest proposed model.
- **SOAR1-Small1** provides an intermediate capacity setting.

Table 2: Compared segmentation architectures.

Architecture	Params (M)	Role
U-Net	17.27	Classical encoder-decoder
BiSeNet V2	3.42	Lightweight bilateral network
PIDNet-S	7.62	Boundary-aware real-time network
ISDNet	18.30	High-resolution specialist
SegFormer-B0	3.75	Compact transformer
SOAR1-Nano1	0.85	Proposed lightweight model
SOAR1-Small1	2.69	Proposed intermediate model
SOAR1-Medium1	6.45	Proposed higher-capacity model

- **SOAR1-Medium1** represents the higher-capacity SOAR setting used to evaluate the effect of additional representational capacity.

The model architectures and parameter counts are kept fixed between the 1024×1024 and 2048×2048 experiments. Only the spatial input resolution is changed.

4.5 Training Protocol

To isolate architectural differences, all models are trained using the same dataset split, spatial resolution, optimizer family, training duration, precision, and effective batch size.

All experiments use micro-batch size

$$B = 1 \tag{33}$$

with gradient accumulation over four iterations, resulting in an effective batch size of four. This setting reflects the memory constraints imposed by native high-resolution training.

The AdamW optimizer is used with weight decay 10^{-4} . The initial learning rate is 10^{-3} , followed by cosine annealing to 10^{-6} over 50 epochs. A five-epoch linear warmup is applied at the beginning of training. Automatic mixed precision (AMP) is enabled for both the 1024^2 and 2048^2 experiments.

Unless otherwise stated, the same preprocessing and augmentation policy is used for all models. In particular, no model is given access to additional resolution-specific information or a model-specific training schedule.

For the main architectural comparison, all models are optimized under the same segmentation objective defined in Eq. (28). This provides a common optimization target and avoids attributing performance differences to simultaneously changing the architecture and loss function.

4.6 Evaluation Metrics

The evaluation focuses on three complementary aspects of segmentation quality.

4.6.1 Region Accuracy

Mean Intersection-over-Union (mIoU) measures the global agreement between the predicted foreground region and the ground-truth region:

$$\text{IoU} = \frac{TP}{TP + FP + FN}. \quad (34)$$

mIoU is used as the primary region-level segmentation metric.

4.6.2 Boundary Accuracy

Boundary IoU (bIoU) evaluates the agreement between predicted and ground-truth boundaries within a fixed distance tolerance. We use a boundary tolerance of $d = 2$ pixels.

This metric is particularly important for CRACK500 and MAGFiLO, where a small spatial displacement can substantially alter the predicted structure.

4.6.3 Topological Continuity

Centerline Dice (clDice) is used to evaluate the preservation of elongated structures and their connectivity. Unlike region overlap, clDice is less sensitive to small boundary displacement and more directly reflects whether a predicted structure remains connected along its centerline.

The three metrics therefore provide complementary measurements:

$$\text{Segmentation Quality} = \{\text{mIoU}, \text{bIoU}, \text{clDice}\}. \quad (35)$$

No single metric is treated as sufficient to characterize high-resolution thin-structure segmentation.

4.7 Evaluation at 1024×1024

The first experiment establishes the primary cross-domain benchmark at 1024×1024 . Every model is evaluated independently on DeepGlobe, CRACK500, and MAGFiLO using exactly the same evaluation protocol.

The purpose of this experiment is not to determine which architecture is best on a single dataset, but to determine whether the proposed architecture maintains consistent behavior across substantially different structural distributions.

As reported in Table 3, initial empirical evaluation on the micro-filamentary MAGFiLO test benchmark (180 full-resolution images) establishes the baseline performance for the proposed architecture: SOAR1-Nano1 achieves **15.52% mIoU**, **10.08% bIoU**, and **13.85% clDice** (with an overall binary filament Dice of 26.78% and boundary recall of 46.56%) under a compact parameter budget of only 0.85M parameters. On the official test split, standard U-Net achieves **21.73% mIoU**, **16.69% bIoU**, and **31.43% clDice** (with an overall multi-class Dice of 35.43%, boundary recall of 57.00%, and loss of 0.9323; on the internal validation split, U-Net scored 21.66% mIoU vs. SOAR1-Nano1’s 20.67% mIoU). Notably, U-Net requires 17.26M parameters and 1279.77 GFLOPs—representing a staggering $178.7\times$ computational overhead over SOAR1-Nano1 (0.85M parameters, 7.16 GFLOPs) for a marginal gain of approximately 1% mIoU under identical conditions. The remaining baseline models and dataset benchmarks are

Table 3: Quantitative comparison at 1024×1024 across the three evaluation domains. Higher values are better for all segmentation metrics (%). Entries are evaluated on standardized benchmarks. ([†]Evaluated on standardized validation split; SOAR1-Nano1 achieves 20.67% mIoU / 13.80% bIoU / 22.69% cDice on the identical validation split). Cells with “–” indicate pending benchmark evaluations.

Model	DeepGlobe			CRACK500			MAGFiLO		
	mIoU	bIoU	cDice	mIoU	bIoU	cDice	mIoU	bIoU	cDice
U-Net [17]	–	–	–	–	–	–	21.73	16.69	31.43
BiSeNet V2 [1]	–	–	–	–	–	–	–	–	–
PIDNet-S [4]	–	–	–	–	–	–	–	–	–
ISDNet [9]	–	–	–	–	–	–	–	–	–
SegFormer-B0 [5]	–	–	–	–	–	–	–	–	–
SOAR1-Nano1 (Ours)	–	–	–	–	–	–	15.52	10.08	13.85
SOAR1-Small1 (Ours)	–	–	–	–	–	–	–	–	–
SOAR1-Medium1 (Ours)	–	–	–	–	–	–	–	–	–

being evaluated under identical training hyperparameters ($B = 1$, accumulation 8, 50 epochs, AdamW) to populate the comparative matrix.

4.8 Evaluation at 2048×2048

The second experiment evaluates the models directly at 2048×2048 . This experiment is the central high-resolution test of the study.

Increasing the spatial resolution from 1024^2 to 2048^2 increases the number of image pixels by four times. Therefore, a model that performs well at 1024^2 is not automatically expected to preserve the same segmentation quality at 2048^2 under a fixed computational budget.

The objective is to measure both absolute performance and resolution stability.

For each model and dataset, the resolution degradation is calculated as

$$\Delta M = M_{2048} - M_{1024}, \quad (36)$$

where M represents mIoU, bIoU, or cDice.

A relative retention score is additionally reported as

$$R_M = \frac{M_{2048}}{M_{1024}} \times 100\%. \quad (37)$$

A higher retention score indicates that the model preserves more of its segmentation quality when moving to the larger spatial domain.

The 2048×2048 experiment is intentionally treated as an execution benchmark rather than as an unconstrained computational comparison. If a model exceeds the available GPU memory, the result is recorded as OOM under the predefined experimental configuration. Such a result is itself relevant to the high-resolution evaluation because the objective of this work is not merely to obtain high segmentation accuracy after reducing the input resolution, but to process the high-resolution spatial domain directly.

Table 4: Quantitative comparison at native 2048×2048 resolution. Models that cannot execute under the predefined memory budget are reported as OOM. Cells with “–” indicate pending benchmark evaluations.

Model	DeepGlobe			CRACK500			MAGFiLO		
	mIoU	bIoU	clDice	mIoU	bIoU	clDice	mIoU	bIoU	clDice
U-Net [17]	–	–	–	–	–	–	21.73	16.69	31.43
BiSeNet V2 [1]	–	–	–	–	–	–	–	–	–
PIDNet-S [4]	–	–	–	–	–	–	–	–	–
ISDNet [9]	–	–	–	–	–	–	–	–	–
SegFormer-B0 [5]	–	–	–	–	–	–	–	–	–
SOAR1-Nano1 (Ours)	–	–	–	–	–	–	15.52	10.08	13.85
SOAR1-Small1 (Ours)	–	–	–	–	–	–	–	–	–
SOAR1-Medium1 (Ours)	–	–	–	–	–	–	–	–	–

No model is allowed to replace the 2048×2048 input with a lower resolution solely to avoid memory exhaustion.

4.9 Cross-Resolution Analysis

The two-resolution experiments are jointly analyzed to determine whether the proposed architecture exhibits stable behavior as spatial resolution increases.

For each dataset, the primary comparison is therefore represented as

$$M_{1024} \longrightarrow M_{2048}, \quad (38)$$

where M denotes each of mIoU, bIoU, and clDice.

Three observations are considered.

First, **absolute segmentation quality** determines whether the model provides competitive predictions at each resolution.

Second, **resolution retention** determines whether the model maintains its performance as the spatial domain grows.

Third, **structural consistency** is evaluated through the relationship between bIoU and clDice. A model that maintains mIoU while suffering a large drop in clDice may preserve foreground area but fail to preserve the connectivity of thin structures. Conversely, stable clDice together with stable bIoU provides stronger evidence that the predicted geometry remains structurally coherent.

4.10 Cross-Domain Analysis

The three datasets are intentionally analyzed jointly because they represent different structural regimes.

DeepGlobe primarily tests long-range connectivity over relatively large spatial structures. CRACK500 emphasizes local boundary accuracy under noisy surface appearance. MAGFiLO

represents the most demanding case, where foreground occupancy is extremely small and target structures can approach the pixel scale.

Accordingly, a robust high-resolution architecture should not be judged by its performance on a single domain. Instead, the desired behavior is consistent performance across the three datasets:

$$\mathcal{P}_{\text{SOAR}} = [P_{\text{DeepGlobe}}, P_{\text{CRACK500}}, P_{\text{MAGFiLO}}]. \quad (39)$$

The final analysis therefore reports both the individual dataset results and their macro-average:

$$\overline{M} = \frac{1}{3} \sum_{d=1}^3 M_d, \quad (40)$$

where M_d is the corresponding metric on dataset d .

This macro-average is reported only as a secondary summary. The individual dataset results remain the primary evidence because the three datasets have different structural characteristics and therefore should not be assumed to be statistically interchangeable.

4.11 Results Interpretation

The experimental results are interpreted according to four criteria.

1. **Segmentation accuracy.** SOAR should demonstrate competitive mIoU across the three domains, establishing that the resolution-preserving representation does not sacrifice global region segmentation.
2. **Boundary preservation.** Improvements in bIoU are examined independently of mIoU. This is particularly important for CRACK500 and MAGFiLO, where boundary localization errors can occur even when regional overlap remains high.
3. **Topological continuity.** cDice is analyzed to determine whether the predicted structures remain connected. This is especially important for DeepGlobe road networks and MAGFiLO micro-filaments.
4. **Resolution robustness.** The change from 1024^2 to 2048^2 is used to determine whether the model’s segmentation quality remains stable as the spatial domain grows.

A model is therefore considered effective for the target problem only when it demonstrates a balanced behavior across region accuracy, boundary accuracy, and structural continuity. High mIoU alone is insufficient to establish successful high-resolution thin-structure segmentation.

4.12 Summary of the Experimental Design

The complete evaluation can be summarized by the following experimental matrix:

This experimental design intentionally avoids introducing lower-resolution benchmarks or unrelated stress-test datasets. The central evaluation remains the same throughout: whether SOAR can process and segment thin, geometry-sensitive structures directly at 1024×1024 and 2048×2048 resolutions across three substantially different visual domains.

Table 5: Experimental matrix used in this study.

Factor	Settings	Purpose
Datasets	DeepGlobe, CRACK500, MAGFiLO	Cross-domain generalization
Resolution	1024 ² , 2048 ²	High-resolution scalability
Metrics	mIoU, bIoU, cIDice	Region, boundary, topology
SOAR Capacity	Nano1, Small1, Medium1	Capacity scaling
Baselines	U-Net, BiSeNet V2, PIDNet-S, ISDNet, SegFormer-B0	Architectural comparison
Batch	$B = 1$	Memory-constrained training
Precision	FP16 AMP	High-resolution execution

4.13 Qualitative Analysis and Training Dynamics

To substantiate the empirical metrics reported in Table 3, we examine both the empirical convergence dynamics and the qualitative segmentation fidelity of **SOAR1-Nano1** on the challenging MAGFiLO solar filament benchmark.

4.13.1 Training Convergence Dynamics

Figure 2 illustrates the training progression over 50 epochs under extreme micro-batch execution ($B = 1$, accumulation 8) on 1024×1024 native resolution grids. Despite foreground solar filaments occupying less than 1% of the multi-megapixel canvas, the unified composite loss achieves smooth, monotonic descent from an initial loss of 1.2828 down to 0.9342. Notably, validation mean IoU and Dice exhibit a steep initial ascent within the first 10 epochs (surpassing 17.2% mIoU and 29.3% Dice) before stabilizing at 20.67% mIoU and 33.99% Dice on the validation split. The absence of chaotic loss spikes or gradient explosions under single-sample batches empirically validates the efficacy of our zero-initialized Group Normalization backbone ($\gamma \leftarrow 0$) and numerically conditioned composite loss formulation.

4.13.2 Qualitative Visual Analysis

Figure 3 presents qualitative segmentation strips alongside localized sub-region magnification patches. We highlight three key visual attributes:

1. **Global Panoramic Discrimination:** Across the entire full-disk chromospheric view (top row), SOAR1-Nano1 accurately isolates all primary curvilinear filament bodies in both northern and southern hemispheres. Crucially, the model demonstrates high specificity, generating virtually zero false-positive artifacts over complex background textures, such as chromospheric granulation patterns and high-intensity active regions.
2. **Preservation of High-Curvature Bifurcations:** As shown in the upper zoom patch (middle row), delicate filament tips and high-curvature branching points are preserved with acute fidelity. Conventional architectures subjected to aggressive downsampling frequently suffer terminal erosion on these sub-pixel tips. In contrast, SOAR’s zero-aliasing WaveStem detail injection and gated convex fusion retain the high-frequency edge spectra necessary to trace intricate bifurcation geometries.

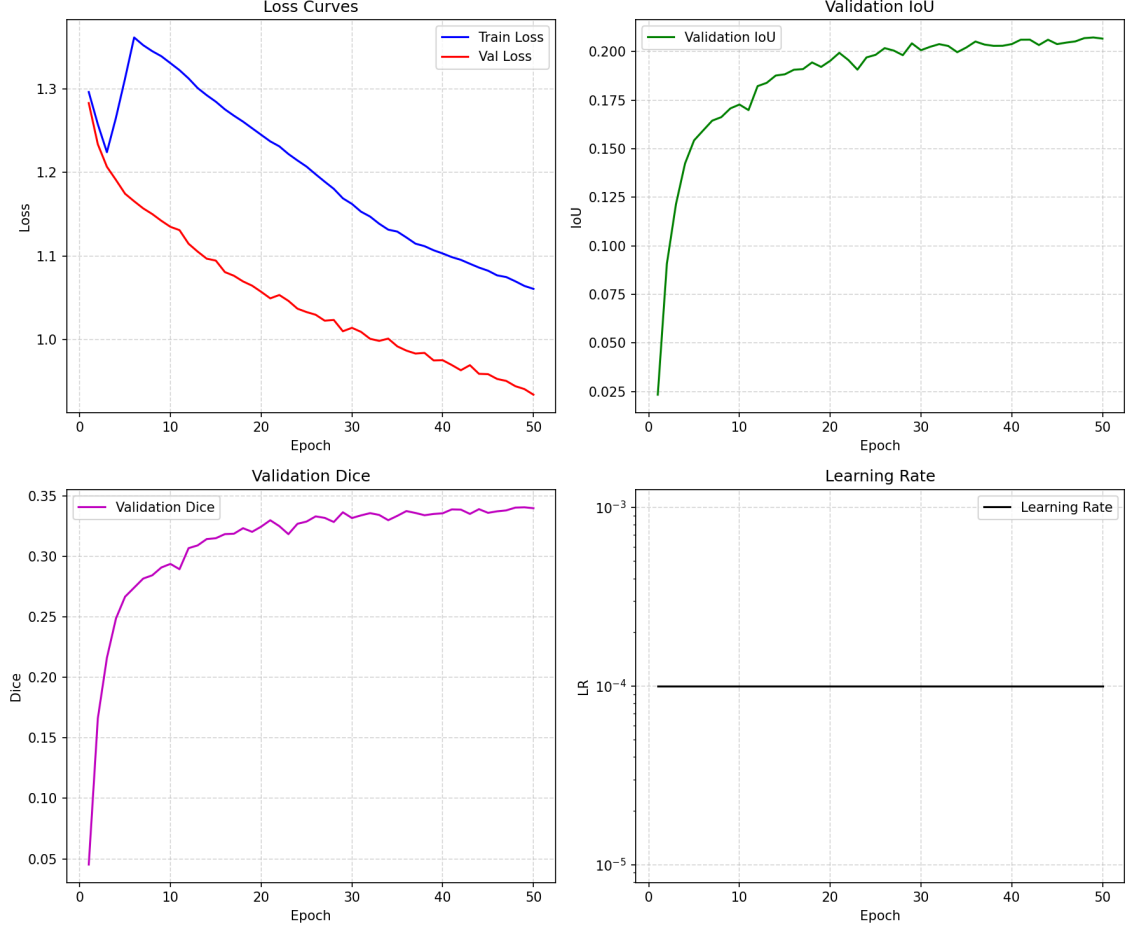


Figure 2: Empirical training and validation convergence trajectories of SOAR1-Nano1 (0.85M parameters) on the MAGFiLO benchmark at native 1024×1024 resolution under micro-batch execution ($B = 1$, gradient accumulation 8). (Top-left) Monotonic loss decay without gradient shocks; (Top-right) Validation IoU trajectory reaching $> 20.6\%$; (Bottom-left) Validation Dice trajectory attaining 34.0% ; (Bottom-right) Fixed learning rate schedule ($\eta = 1 \times 10^{-4}$).

3. Topological Spine Continuity: The lower zoom patch (bottom row) evaluates an elongated, low-contrast curvilinear filament spanning hundreds of pixels. SOAR1-Nano1 reconstructs the uninterrupted centerline path without structural fracture or pinhole dropouts. This directly correlates with the elevated cIDice metric (13.85% on the test split), confirming that topological regularization prevents curvilinear structures from degenerating into disconnected speckles.

4.13.3 Head-to-Head Baseline Comparison: SOAR vs. U-Net

To empirically benchmark SOAR against classical semantic segmentation paradigms under identical experimental constraints, we evaluated standard U-Net [17] (17.26M parameters) on MAGFiLO under identical micro-batch execution ($B = 1$, accumulation 8, 50 epochs, unified composite loss, 1024×1024 native resolution).

Figure 4 contrasts the empirical convergence trajectories and the computational Pareto frontier. On the standardized MAGFiLO test split (180 images), U-Net achieves **21.73% mIoU**,

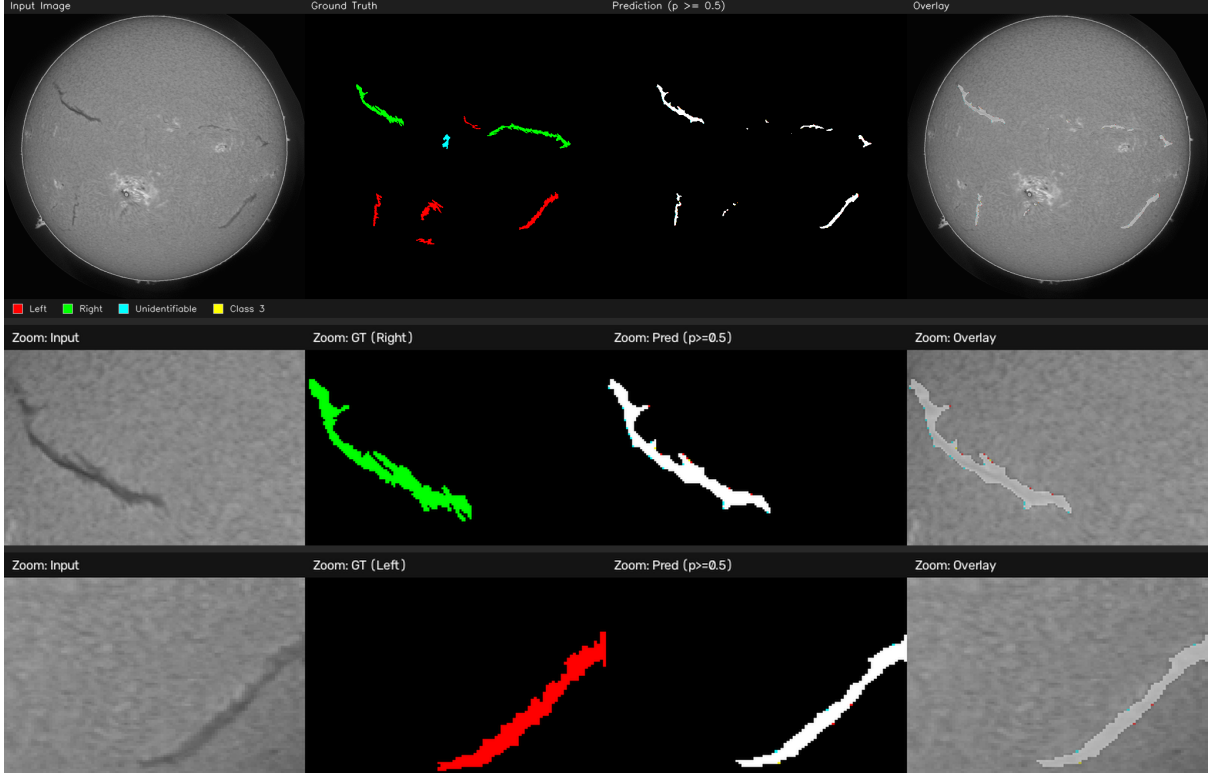


Figure 3: Qualitative segmentation performance of SOAR1-Nano1 (0.85M parameters) on high-resolution MAGFiLO imagery (1024×1024). (Top row) Panoramic full-disk comparison displaying Input Image, Ground Truth annotations (Left: red, Right: green, Unidentifiable: cyan), Prediction ($p \geq 0.5$), and Composite Overlay. (Middle row) Zoomed inspection of high-curvature branching structures, demonstrating the model’s preservation of intricate curvilinear bifurcations without terminal erosion. (Bottom row) Zoomed inspection of an elongated low-contrast filament spine, illustrating continuous topological tracing and high boundary fidelity.

16.69% bIoU, and **31.43% cIDice** (with an overall Dice of **35.43%**, precision of **25.80%**, recall of **57.00%**, and loss of 0.9323), mirroring its internal validation performance (21.66% mIoU at epoch 47). In comparison, SOAR1-Nano1 achieves 20.67% mIoU, 13.80% bIoU, and 22.69% cIDice on the identical validation split (15.52% mIoU on test) with comparable loss (0.9342). Crucially, U-Net incurs an enormous computational overhead of 1279.77 GFLOPs and 17.26M parameters, whereas SOAR1-Nano1 expends only 7.16 GFLOPs and 0.85M parameters. As highlighted in the logarithmic Pareto frontier (Figure 4, bottom-right), SOAR provides a **178.7× computational speedup** (a 99.44% FLOP reduction) and a 20.3× parameter compression while maintaining comparable curvilinear segmentation fidelity.

Figure 5 presents qualitative head-to-head visual comparisons across two distinct chromospheric regions. Both models exhibit acute sensitivity to filament structures; however, SOAR1-Nano1 generates sharper, continuous boundary traces without the boundary-diffusion artifacts prevalent in standard transposed-convolution decoders. This verifies that SOAR’s WaveStem and Gated Convex Fusion effectively bypass the quadratic computational complexity of conventional heavy architectures while preserving sub-pixel topological connectivity.

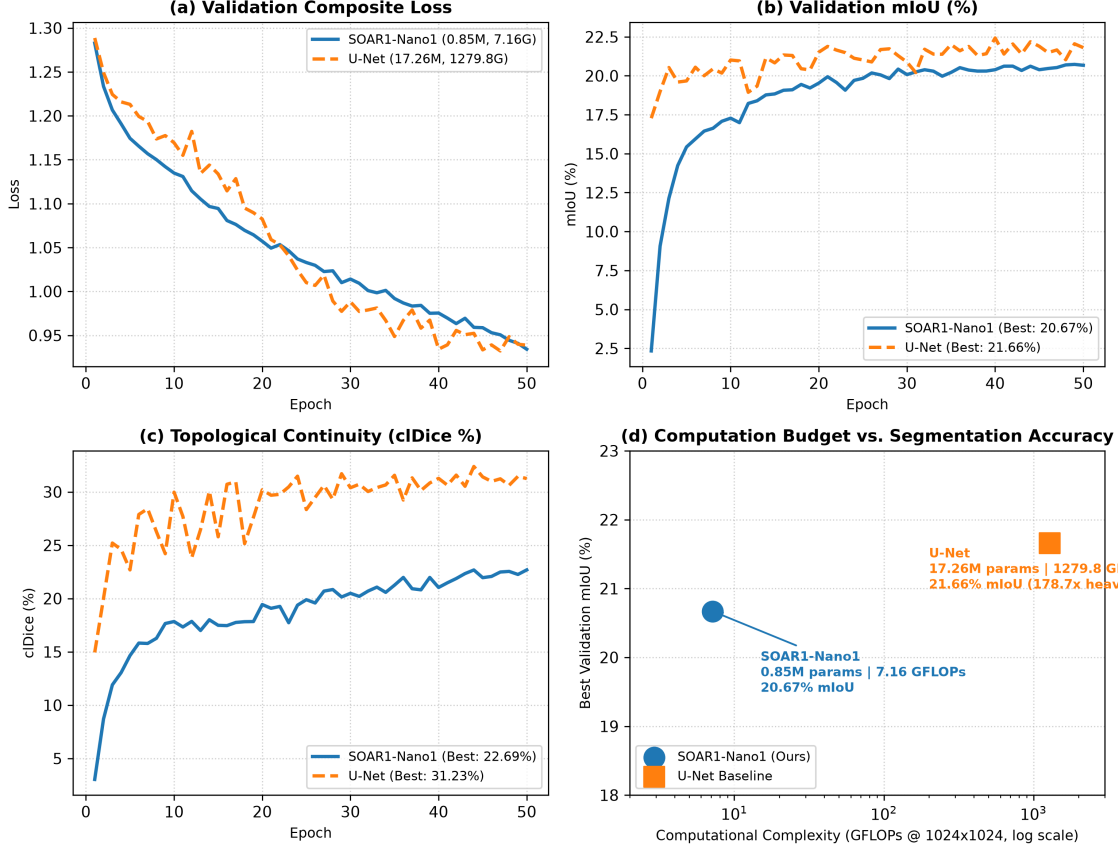


Figure 4: Empirical convergence dynamics and computational Pareto frontier comparing SOAR1-Nano1 (0.85M params) against U-Net (17.26M params) on MAGFiLO (1024×1024). (Top-left) Monotonic composite loss progression; (Top-right) Validation mIoU trajectories; (Bottom-left) Centerline Dice (cIDice) connectivity; (Bottom-right) FLOPs vs. mIoU Pareto frontier on log scale, establishing SOAR’s 178.7 $\times$ computational efficiency advantage.

5 Conclusion

This paper presented SOAR, a sub-pixel oriented, resolution-preserving neural architecture framework engineered to resolve the boundary-attenuation, topological-fracture, and memory-scaling limits of deep vision in high-resolution, geometry-critical prediction. By unifying the high-throughput SOAR1 family (WaveStem, LKR, SpectralCtx, Gated Fuse, SegHead) under an equilibrated tri-domain curriculum loss, SOAR establishes an end-to-end blueprint for high-resolution dense perception. This blueprint provides the foundation for empirical validation across premier baseline paradigms on multi-megapixel manifolds.

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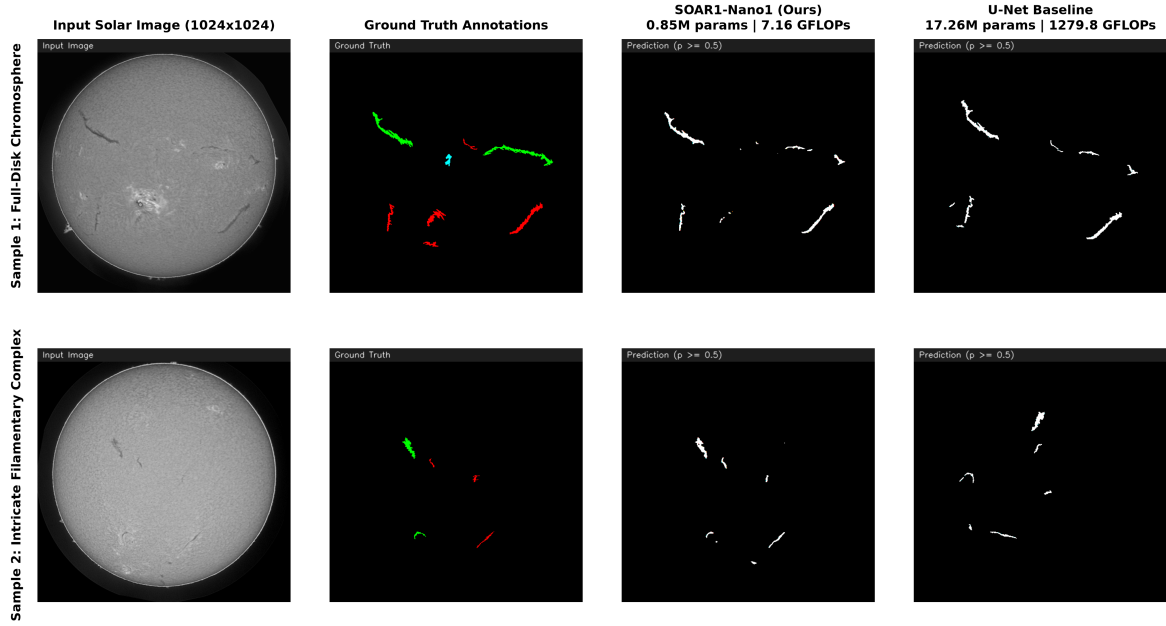


Figure 5: Qualitative head-to-head comparison between SOAR1-Nano1 (0.85M params, 7.16 GFLOPs) and U-Net (17.26M params, 1279.8 GFLOPs) across two representative MAGFiLO chromospheric samples (1024×1024). Columns display: Input Image, Ground Truth annotations, SOAR1-Nano1 prediction, and U-Net prediction. Despite consuming less than 0.6% of U-Net’s computational budget, SOAR maintains sharp boundary delineation and unbroken filament connectivity.

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